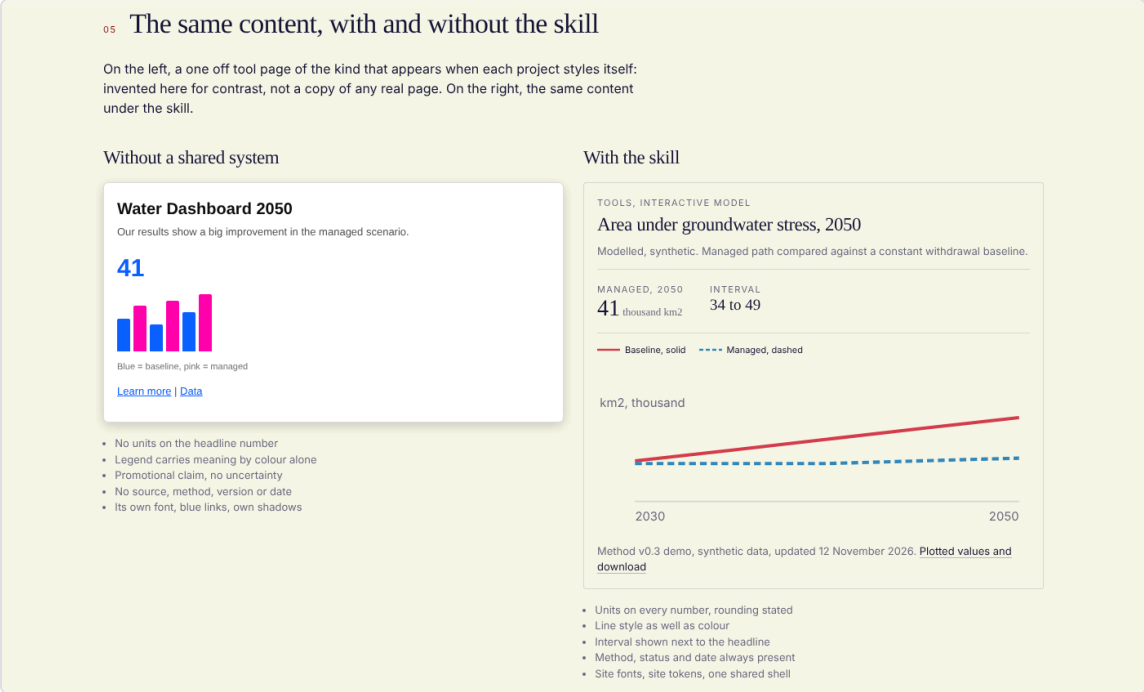
Sustainable Solutions Lab

STANFORD SUSTAINABLE SOLUTIONS LAB, RESEARCH TOOLS

A design skill for the lab's tools, and what it produces

A design system for `sustainablesolutions.stanford.edu/tools`, written so that Claude Code applies it while building a tool. It contains the site's verified design tokens, the order of sections on a tool page, the states each component must define, and the attribution a research tool has to carry. This guide shows the rules and the resulting output.



SKILL	RUNS IN	TOKENS VERIFIED	GUIDE ISSUED
sustainable-solutions-ui, v2.1.0	Claude Code, project scoped	9 November 2026	12 November 2026

Live version of these pages: `claude.ai/artifact/RzcxrbJUQiTKZGmMSZ4Nie`
Interactive: toggle dark mode, tab through the controls, resize to phone width.

Every figure in this guide is a real screenshot of a page built with the skill. All data, regions, papers and figures in those screenshots are synthetic and must not be read as lab results.

01

What this is, and the problem it solves

Tools are added to the site one at a time, by different people, over months.

The problem

Each tool is built by a different person at a different time. Without a shared reference, fonts, colours, spacing and link styles differ between tools, legends end up using colour as the only key, and the data source, version and method are not always shown next to the figure.

A written style guide does not prevent this, because it is usually not open at the moment the code is written.

The approach

The rules are written for the agent that writes the code. The skill sits in the repository and loads when a task concerns a tool page. It contains four things:

- design tokens extracted from the live site, each with a verification status
- a mandatory page skeleton for a tool page
- component state coverage, from focus ring to empty result
- science and attribution rules: units, uncertainty, provenance, credits

Values that must not be invented

SKILL.md lists values that must be taken from the live site or the repository and never guessed: hex codes, font implementation, grid measurements, file paths, institutional endorsements, exact data, source credits and study results. Where a value cannot be confirmed, it is labelled as a proposal.

> Make every research tool feel like a native part of a single laboratory website while preserving scientific meaning, transparency, accessibility, and performance.

The skill's stated mission.

Evidence order

When two sources disagree, the skill sets the order: the live application source first, then sibling tool pages already on the site, then its own browser verified audit, then Stanford identity guidance as an institutional baseline, and only last its own proposed defaults, which must be labelled as proposals.

> Version 1 of the skill was written from public text alone and recorded the wrong font, background colour and link colour. Version 2 was rebuilt from browser inspection of the live site on 9 November 2026. Each corrected value is listed in the changelog.

Where it applies

PAGE TYPE	WHAT THE USER MUST BE ABLE TO DO
Catalog, /tools	Identify and enter the right tool
Tool landing	Understand the research question before touching a control
Interactive model	Change inputs and interpret the output, with uncertainty
Interactive map	Find and compare spatial patterns, without gestures being the only route
Plot or dashboard	Compare variables and scenarios on consistent scales
Dataset page	See provenance and reuse records safely
Media essay	Learn from images and video with captions, transcripts and rights

02

Design tokens

These are the production values, read from `src/styles/colors and type.css` and from the computed styles of the live site on 9 November 2026. Tool code references the token name, not the literal value.

Surfaces and ink

`--paper` `#f8f8e8`	`--paper-2` `#f1f1df`	`--paper-3` `#e8e8d4`	`--ink` `#181838`	`--ink-2` `#3a3a5a`	`--ink-3` `#6b6b80`

`--cardinal` `#8c1515`	`--brand-teal` `#78c8d8`	`--brand-green` `#48a848`	`--brand-orange` `#e87828`

The background is cream, not white, and the text is navy, not black. Links use the text colour with a bottom border instead of a separate link colour. In dark mode paper becomes `#0c0c1c`, ink becomes the cream, and the accent lightens to `#e04545`.

Data palette



Eleven stop spectral diverging ramp. The default two series pair is `--spectral-10` blue and `--spectral-2` red. Colour is not used as the only differentiator: line style, marker shape or a direct label carries the same information.

Type

Sans	Inter, 400 to 700, Google Fonts
Serif	Source Serif 4 renders today. Tiempos Headline is first in the stack but not licensed.
Code	ui-monospace, SF Mono, Menlo, Consolas
Scale	11, 13, 16, 19, 23, 28, 34, 41, 56, 72 px
Line height	1.12 tight, 1.28 snug, 1.5 normal, 1.7 loose
Eyebrow	11px, uppercase, 0.12em tracking

Layout, radius, motion

Spacing	4px base: 4, 8, 12, 16, 24, 32, 48, 64, 96, 128
Container	1200px
Prose measure	680px
Header	57px, sticky, 1px rule under
Tool frame	calc(100dvh minus 57px)
Radius	2, 4, 8 px
Motion	120ms control, 200ms interface, 400ms content, one easing curve, all disabled under reduced motion

Research domain colours, with measured contrast on cream

Energy #c2440e, 4.75:1

Food #2e7b3f, 4.87:1

Water #1f5e8c, 6.45:1

Materials #5e4fa2, 6.28:1

Climate #1f6b82, 5.62:1

Health #9e0142, 7.72:1

Each domain colour was measured against the cream background. The skill includes a validator script that checks token structure, contrast and completeness when the tokens change.

03

The tokens, rendered

The foundations section of the showcase page. Every value shown is read from the token file.



Screenshot, light mode, desktop width.

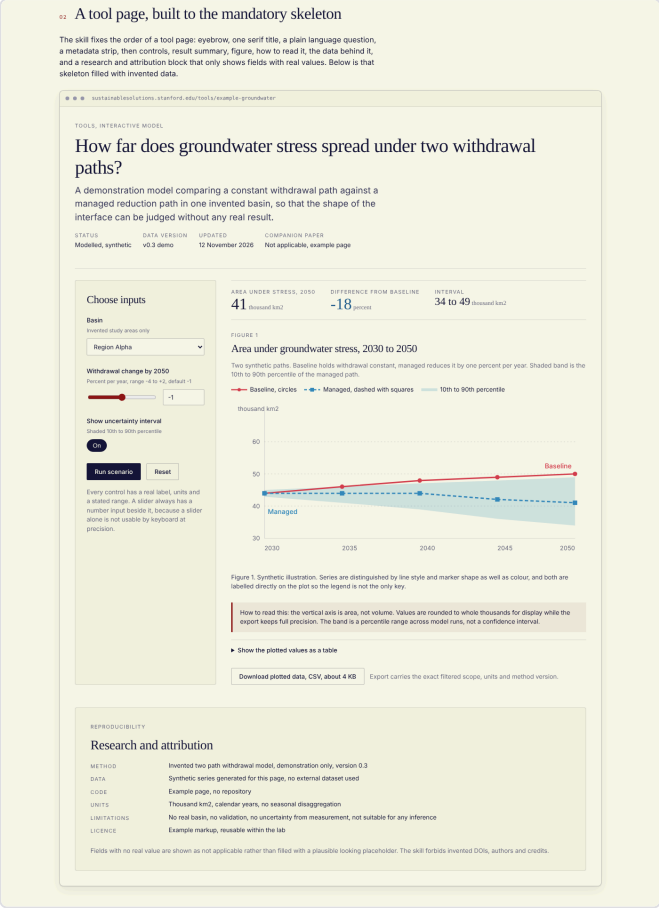
04

The page skeleton

Every tool page uses the same order of sections, so the same information is in the same place on each tool.

1. Shared masthead and skip link. No tool invents its own header.
2. Eyebrow, single serif h1, one paragraph stating the scientific question in plain English.
3. Metadata strip: status, data version, updated date, companion paper, only where a real value exists.
4. Controls with labels, units, ranges and defaults.
5. Result summary before the figure, so the answer is readable without interpreting a plot.
6. Figure with legend, then how to read this.
7. The plotted values as a table, and a download that carries the exact filtered scope.
8. Methods, uncertainty, scope and limitations.
9. Research and attribution: paper, data, code, units, licence, credits, footnotes.
10. Shared site footer.

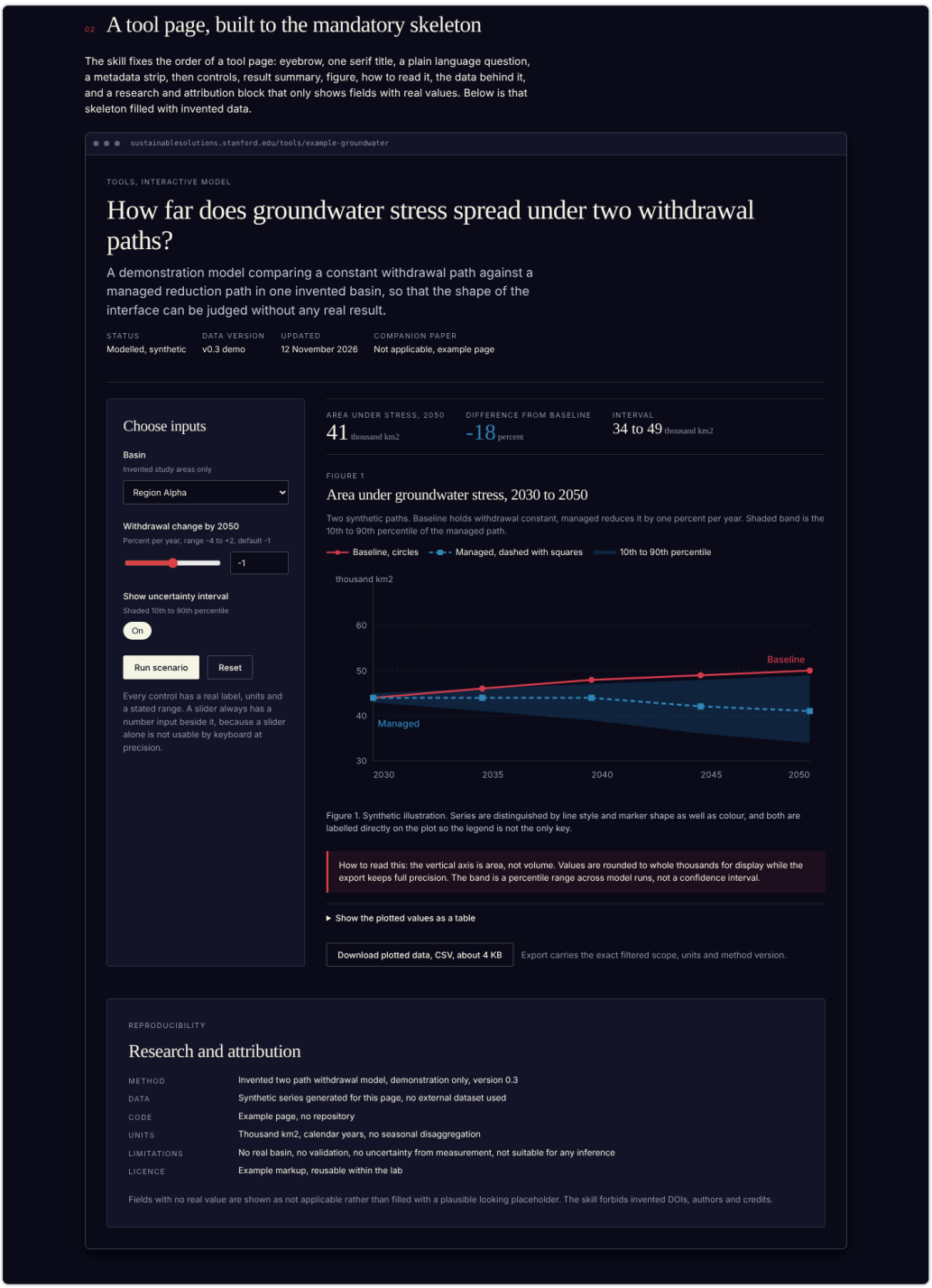
> Fields without a real value are shown as not applicable. Placeholder DOIs, authors and credits are not written, because a placeholder left in production reads as a statement of fact.



05

Dark mode

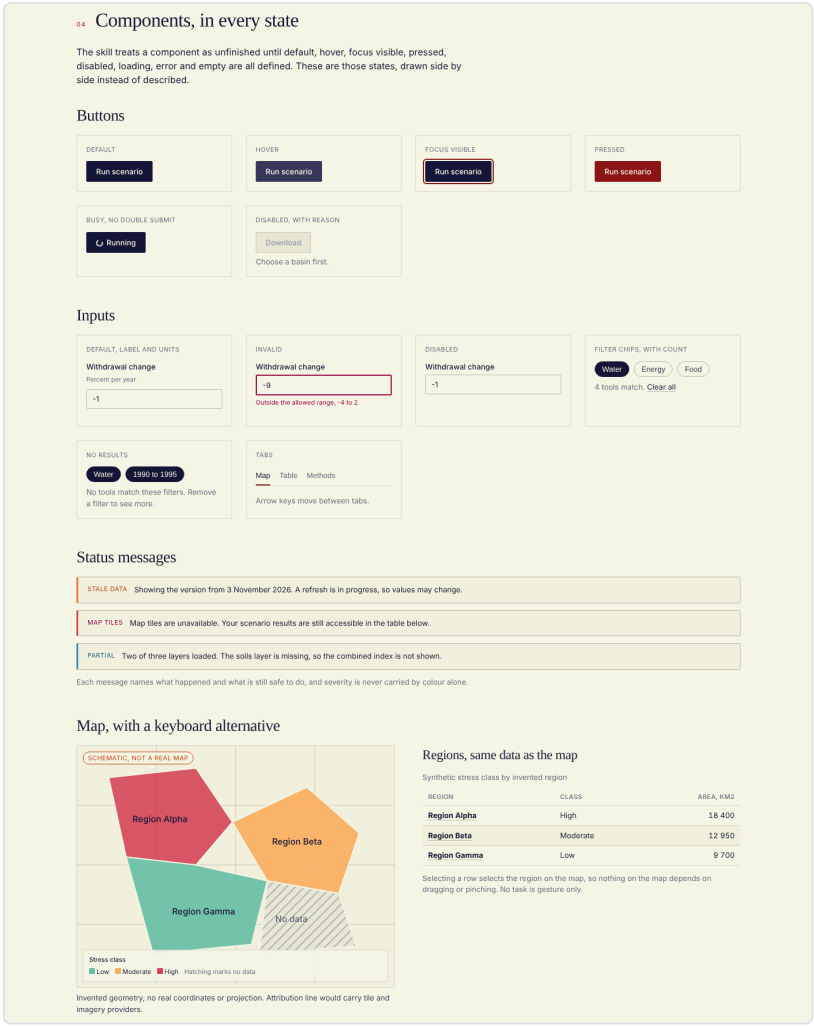
The same page and the same markup, with the token values swapped. The theme writes `data-theme` on the html element and remembers the choice under `localStorage, key ssl-theme`, taking the operating system preference as the first default. Dark mode changes paper, ink and the accent, and the values derived from them follow.



06

Component states

A component is complete when default, hover, focus visible, pressed, disabled with a reason, loading, error, empty and the narrow viewport are all defined. The skill includes a state matrix to fill in per component. The screenshot below shows those states rendered.



RULE	REASON
No colour only, hover only or gesture only meaning	Keyboard users, colour vision differences, touch devices
A slider always has a number input beside it	A slider alone cannot be set precisely by keyboard
Disabled controls state the reason	A disabled control with no explanation is read as a fault
Error messages say what happened and what is still safe to do	Partial failure is the normal case in data tools
Every map has a list or table alternative	No task may require dragging or pinching

07

Rules for research content

Figures

- Axes carry units and the scale is named. Baselines and scenarios are stated, not implied.
- Display rounding is separated from computation and export precision.
- Uncertainty intervals are labelled for what they are. A percentile band is not a confidence interval.
- Every essential plotted value is reachable as a table.
- Series are distinguished by line style or marker as well as colour, and labelled directly where space allows.

Maps

- Legend, bounds, projection and spatial metadata are present.
- Selection works from a list as well as from the map.
- Tile and imagery providers are credited.
- A tile failure still leaves the results readable.

Attribution

- Paper, authors, venue and year, with a real DOI or nothing.
- Dataset with version or date, method, code repository and licence.
- Image, video and cartography credits, alt text, captions or transcript.
- Footnotes with stable anchors that survive print and export.

Copy

- Plain, neutral, research first. Never promotional.
- Observation, model output and projection are named as such.
- Descriptive labels: download plotted data, CSV. Not: data.
- Honest status: estimated, modelled, observed, updated on a date, not available, outside the study area.

> Definition of done: every number on the page can be traced to a dataset and a method, the workflow can be completed by keyboard and on a narrow screen, and no placeholder remains.

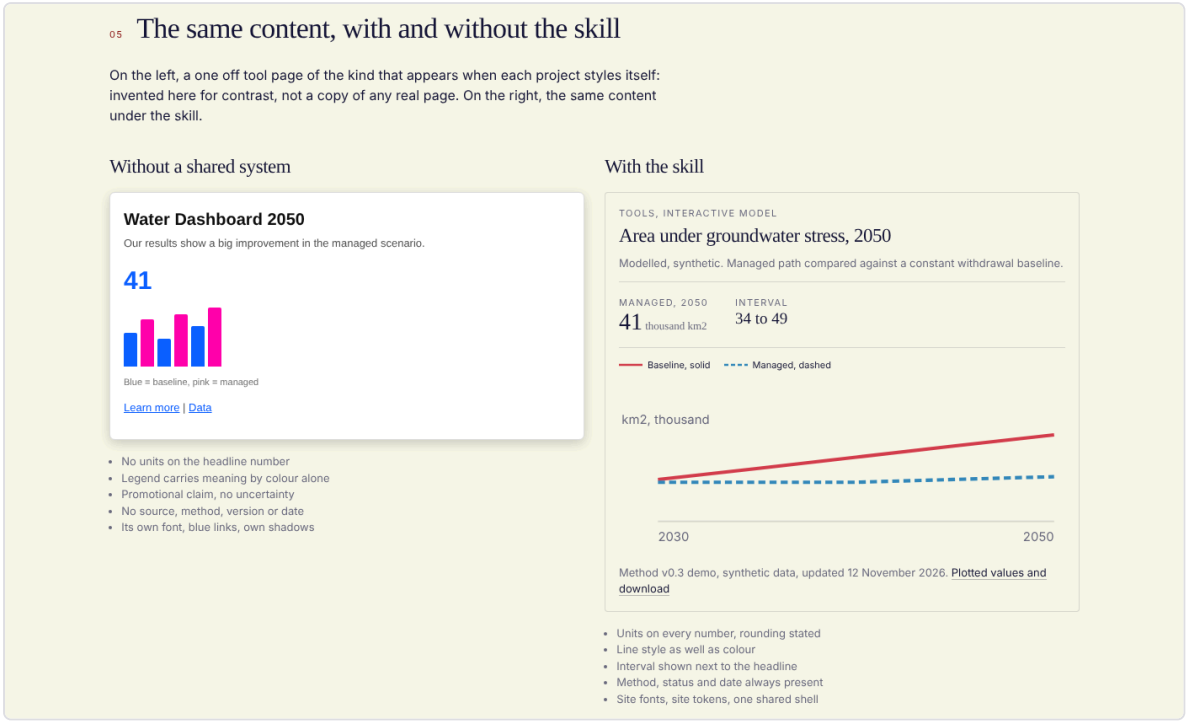
Accessibility and quality gate

- ☐ WCAG 2.2 AA as the target, tested
- ☐ Keyboard path through the whole workflow
- ☐ Screen reader names, roles, values, announcements
- ☐ Text zoom to 200 percent and reflow
- ☐ Contrast measured for text, marks and focus rings
- ☐ Reduced motion respected
- ☐ Light and dark both tested
- ☐ Behaviour with no JavaScript, slow data, invalid input
- ☐ Export and copy keep their attribution
- ☐ Reviewed with the scientist who owns the tool

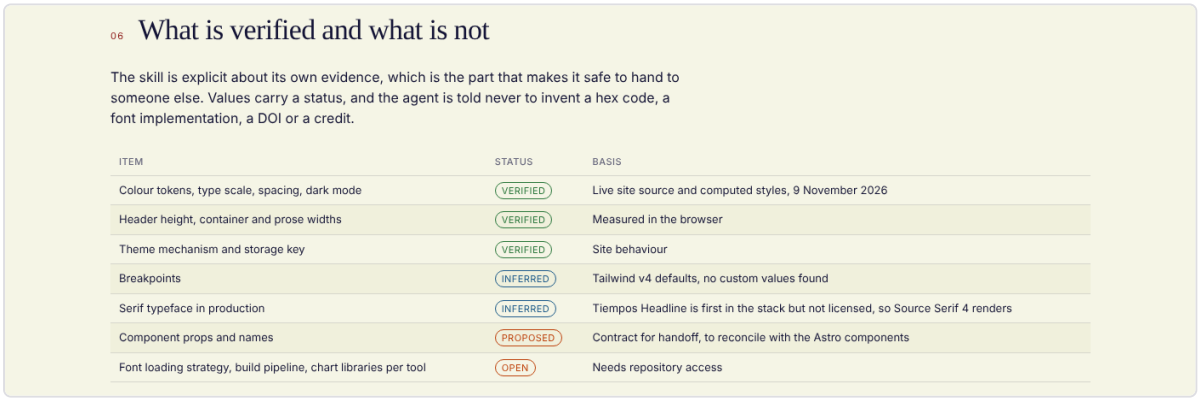
08

Comparison

The same content presented two ways. The left panel was written for this comparison and is not a copy of any existing page.



Verification status



Font loading strategy, the Tiempos Headline licence, the build pipeline, the chart libraries used by each existing tool and the visual regression baselines cannot be confirmed without repository access. They are listed as open.

09

How to use it

Install

1. Copy the folder `.claude/skills/sustainable-solutions-ui/` into the application repository, at the repository root.
2. Open the repository in Claude Code. The skill loads itself when a task mentions the lab site, the tools route, a tool page, a visualization or a UI review.
3. For a deeper pass, give Claude the prompt in `CLAUDE MASTER PROMPT.md` with repository access.

What is in the folder

SKILL.md	Agent instructions, workflow, normative rules
references, 8 files	Site audit, tokens and typography, components and copy, engineering, decisions, charts and maps, media and citations, accessibility
templates, 21 files	Page brief, tool layout, catalog card, forms and states, plot, map, media, citations, provenance, state matrix, review checklists
tokens	CSS and JSON tokens with a JSON schema
examples	Plain HTML and React reference markup
scripts	Token validator, structure, contrast, completeness

Working order for a new tool

1. Design audit of the live page and its siblings
2. Page brief: audience, question, assumptions, primary action
3. Tokens
4. Components
5. Accessible states
6. Plots, maps, media
7. Citations and credits
8. Responsive and assistive testing
9. Release checklist

Suggested governance

One owner for the system. The site owner and the scientist responsible for the tool as approvers. One reviewer who tests with a keyboard and a screen reader. Version the tokens, keep the changelog, and add visual snapshot tests once the skill is in the repository.

What is needed from the lab

- Read access to the site repository, to close the open items on the previous page
- A decision on the serif typeface, since Tiempos Headline is in the stack but not licensed
- One upcoming tool to use as the first test case

Live showcase: claude.ai/artifact/RzcxrbJUQiTKZGmMSZ4Nie. Guide produced 12 November 2026 from the skill as it stands at version 2.1.0. Screenshots are of a demonstration page built with the skill; all data in them is synthetic.